

Rohit Vardhan Kaasa

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Summary:

Quality Engineer with a Master's in Design and Manufacturing and 4 years of proven experience in Quality Engineering and Design Quality Engineering within the Medical Device and Pharmaceutical Manufacturing industries. Skilled in 2D/3D CAD modeling, product design, prototyping, validation, and design assurance, ensuring compliance with ISO 13485, ISO 14971, FDA 21 CFR Part 820, and cGMP regulations. Proficient in SolidWorks, AutoCAD, CATIA, ANSYS, MATLAB, and 3D printing, with demonstrated ability to perform FEA analysis, tolerance stack-up studies, and rapid prototyping. Strong expertise in risk management, FMEA, CAPA, root cause analysis, and continuous improvement initiatives to enhance product quality and patient safety. Recognized for effective collaboration with R&D, Manufacturing, Supplier Quality, and Regulatory Affairs teams, supporting product development from concept through commercialization. Adept at applying Lean Manufacturing and Six Sigma methodologies to streamline processes, reduce defects, and improve efficiency while maintaining compliance in regulated environments.

Skills:

- CAD
- SolidWorks
- AutoCAD
- CATIA
- ANSYS
- MATLAB
- FEA
- FDM
- 3D Printing
- Prototyping
- FMEA
- Root Cause Analysis
- CAPA
- Risk Management (ISO 14971)
- ISO 13485
- FDA
- 21 CFR Part 820
- IEC Standards
- Lean Manufacturing
- Six Sigma
- Continuous Improvement
- MS Office
- Project Management Tools

Professional Experience:

Venture Medical ReQuip, Inc – Tampa, FL | Jan 2025 – Present

Quality Engineer

- Supported drug product development and manufacturing activities by ensuring adherence to FDA, ISO 13485, and cGMP guidelines, thereby maintaining high-quality standards across all phases of the product life cycle.
- Conducted detailed risk assessments (DFMEA, PFMEA, Hazard Analysis), identifying potential design and process failures and implementing risk mitigation strategies to ensure compliance and product reliability.
- Partnered with R&D, Regulatory, and Manufacturing teams to provide quality engineering oversight in product design, process development, and commercial manufacturing activities.
- Drafted, reviewed, and approved validation protocols (IQ, OQ, PQ) for equipment and processes, ensuring robust validation strategies that supported compliant scale-up and technology transfer.
- Led CAPA investigations to address product and process nonconformance's, applying root cause analysis and corrective/preventive measures that improved compliance and reduced repeat deviations.
- Contributed to the creation, review, and maintenance of Design History Files (DHF), Device Master Records (DMR), and Device History Records (DHR) in compliance with FDA QSR requirements.
- Oversaw supplier quality activities, including vendor qualification, audits, and component inspections, ensuring that raw materials and purchased parts consistently met quality specifications.
- Participated in internal and external audits, supporting responses to FDA inspections and ISO certification audits by ensuring thorough documentation and timely closure of audit findings.
- Collaborated with manufacturing teams to implement process monitoring systems, SPC charts, and control plans, ensuring sustained process performance and defect prevention.
- Applied lean manufacturing and Six Sigma methodologies to reduce waste, improve cycle time, and enhance manufacturing efficiency while maintaining high-quality outcomes.
- Conducted training sessions for cross-functional teams on quality system procedures, regulatory requirements, and problem-solving tools, enhancing overall compliance culture within the organization.
- Prepared and presented comprehensive quality reports summarizing trends, key metrics, and CAPA effectiveness to senior leadership, supporting data-driven decision-making.

Huwel Life Sciences – Hyderabad, India | Jan 2020 – Jun 2023

Design Quality Engineer

- Responsible for design quality assurance activities for POCT diagnostic devices, ensuring product compliance with ISO 13485, ISO 14971, IEC 60601, and FDA 21 CFR Part 820 throughout the product development cycle.
- Partnered with R&D teams to ensure design controls were implemented effectively, establishing traceability between user needs, design inputs, outputs, verification, and validation activities.
- Performed risk management activities, developing and maintaining Risk Management Files, Hazard Analyses, and DFMEAs, ensuring patient safety and minimizing device-related hazards.
- Managed design verification and validation testing, including functional testing, mechanical testing, and performance evaluations, ensuring devices met intended use and regulatory requirements
- Overseen Root cause analysis investigations for design and manufacturing issues, implementing CAPA actions that improved product quality and reduced recurring failures.
- Supported supplier quality engineering activities by evaluating and qualifying suppliers, performing audits, and reviewing incoming inspection data for critical components and materials.
- Utilized CAD modeling tools (SolidWorks, CATIA, AutoCAD) to design and improve device components, performing tolerance stack-up, stress analysis, and fatigue testing to ensure robust designs.
- Applied ANSYS simulations and MATLAB calculations to validate design performance, enabling data-driven improvements in device reliability and manufacturability.
- Executed rapid prototyping using FDM 3D printing to accelerate design iterations, allowing quicker validation of design concepts and integration into final device development.
- Collaborated with cross-functional teams during design transfer to manufacturing, addressing production challenges, refining assembly processes, and ensuring manufacturability without compromising quality.
- Maintained comprehensive design documentation, change control records, and technical files, supporting successful regulatory submissions and audits.
- Championed continuous improvement initiatives by applying lean and Six Sigma tools to reduce cycle time, improve yields, and enhance the efficiency of development and production processes.

Academic Projects:

- Fused Deposition Modeling – Optimized printing parameters on consumer-grade 3D printers to improve dimensional accuracy and mechanical properties.
- Mini Air Circulator – Designed and manufactured a 3D-printed handheld device for portable air circulation applications.
- Hydraulic Jack Design Improvements – Conducted assembly analysis, performed failure studies, and implemented design modifications for enhanced load capacity.
- Destin Design Services Pvt. Ltd. – Developed a conceptual engineering design startup model focusing on CAD services, product development, and prototyping.
- Quality Engineering & Process Improvement Projects – Applied lean manufacturing and Six Sigma tools to streamline material flow, reduce waste, and improve productivity.
- Project Engineering & Management – Executed projects with focus on cost optimization, vendor coordination, and cross-functional communication.
- Air-Electro Hybrid Vehicle Prototype – Designed and developed a prototype hybrid vehicle integrating mechanical and electrical systems.

Education:

- Masters in Mechanical Engineering - Florida Institute of Technology - Melbourne, FL
- Bachelors in Mechanical Engineering - Vidya Jyothi Institute of Technology – Hyderabad, India